

C Programming Projects



Exploring practical applications and projects using the C programming language



What are C Programming Projects?

C programming projects are programs or tools that generate, plan, and manage various tasks or applications using the C programming language.

Learning & Practice



Projects can assist you in learning and practicing C skills like file handling, command line parsing, and make files.

Integration



Projects can combine many applications into a single executable or library.

Diverse Applications



C projects may be created for various objectives, including financial management, artistic work, and teaching.



Basic C Projects

Beginner Level

Simple projects that can be built with fundamental C programming concepts



Calculator

Simple arithmetic operations using switch cases or if-else statements



Student Record System

File handling for managing student information



Calendar

Date and day tracking using if-else logic and switch-case statements



Phone Book



Unit Converter



Contact management with
data storage capabilities

Converts basic units such
as temperature, currency,
and mass

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Calculator Project



Simple Arithmetic Calculator

You can build a simple calculator with C using switch cases or if-else statements. This calculator takes two operands and an arithmetic operator (+, -, *, /) from the user.



Input Format

The input must be in the format "number1 operator1 number2" (i.e. 2+4)



Expansion Possibilities

You can expand the program to accept more than two operands and one operator by adding logic



Core Functionality

Based on the operator entered by the user, it conducts the computation on the two operands

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Student Record Management System



Student Information Management

Using C language, you can create a student management system that handles students' records (like Student's roll number, Name, Subject, etc.)



Data Storage

It employs files as a database to conduct file handling activities



Core Operations

Add, search, change, and remove entries



Practical Application

Can be handy for schools or colleges that have to store records of thousands of students

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Calendar & Phone Book Projects



Calendar

If you have ever lost track of which day of the week is today or the number of days in



Phone Book

This Phone Book Project generates an external file to permanently store the user's

that particular month, you should build a calendar yourself.



Implementation

We can implement it using simple if-else logic and switch-case statements



Display Function

The display() function is used to display the calendar and can be modified accordingly

data (Name and phone number).



Learning Value

Helps you understand the core concepts of capacity, record keeping, and data structure



Operations

Add, list, edit or alter, look at, and delete data from a record



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Unit Converter & Intermediate Projects



Unit Converter Project

Forgot how to convert degree Fahrenheit to Celsius? Don't worry. We have a solution for you. This unit converter converts basic units such as temperature, currency, and mass.

Intermediate Level

More complex projects that require intermediate C programming skills



Mini Voting System

Online voting platform for organizations



Tic-tac-toe Game

Classic game using 2D arrays



Matrix Calculator

Mathematical operations on matrices



Voting Systems





Mini Voting System

An online voting system is a software platform that enables organizations to conduct votes and elections securely.



Balance

Strikes a balance between ballot security, convenience, and the overall needs of a voting event



Decision Making

Collects input in a systematic and verifiable manner to help make crucial decisions



Voting System

With this voting system, users can enter their preferences and the total votes and leading candidate can be calculated.



Simplicity

It's a straightforward C project that's simple to grasp



Application

Small-scale election efforts can benefit from this



Tic-tac-toe & Matrix Calculator



Tic-tac-toe Game

Tic-tac-toe, also known as noughts and crosses or Xs and Os, is a two-person paper and pencil game in which each player alternates marking squares in a three-by-three grid with an X or an O.



Winning Condition

The winner is the player who successfully places three of their markers in a horizontal, vertical, or diagonal row



Implementation

You can implement this fun game using 2D arrays in the C



Matrix Calculator

Mathematical operations are an everyday part of our life. Matrices are mathematical structures in which integers are arranged in columns and rows.



Operations

Some fundamental matrix operations are performed in this project



Limitation

The project only considers square matrices



Library Management & Electricity Bill Calculator



Library Management System

Library management is a project that manages and preserves electronic book data based on the demands of students.



Electricity Bill Calculator

The Electricity Cost Calculator project is an application-based micro project that predicts the following month's electricity bill based on the appliances or loads used.



Features



Users

Both students and library administrators can use the system



Implementation Files

main.c, searchbook.c, issuebook.c, viewbook.c, and more

All loads' power rating, unit consumed per day, units consumed per month, and total load calculation



Target Users

People who do not have a technical understanding of calculating power bills

Movie Ticket Booking System



Movie Ticket Booking System

The project's goal is to inform a consumer about the MOVIE TICKET BOOKING SYSTEM so that they can order tickets. The project was created with the goal of making the process as simple and quick as possible.



Core Functions

The user can book tickets, cancel tickets, and view all booking records using the system



Main Purpose

To supply various forms of client facilities as well as excellent customer service



Requirements

Should meet nearly all the conditions for reserving a ticket



Advanced C Projects

Advanced Level

Complex projects that require advanced C programming skills



Snakes and Ladders

Ancient Indian board game with minimum jumps calculation



Lexical Analyzer

Translates stream of letters into lexical tokens



Bus Reservation System

Booking bus tickets in advance with seat management



Dino Game

Recreation of the Offline Google Chrome game



Pac-Man Game

Classic game with graphics implementation



Advanced Projects Details



Snakes and Ladders Game

Snakes and ladders, also known as Moksha Patam, is an ancient Indian board game for two or more players that is still considered a worldwide classic today.



Game Mechanics

The dice value can either be provided by the user or it can be generated randomly



Objective

Given a snake and ladder game, write a function that returns the minimum number of jumps to take the top or destination position



Lexical Analyzer

The Lexical Analyzer program translates a stream of individual letters, which are generally grouped as lines, into a stream of lexical tokens.



Main Purpose

To take a C file and generate a sequence of tokens that can be utilized in the next stage of compilation

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Conclusion

C: A Timeless Programming Language

C is clearly not an obsolete programming language, as evidenced by the fact that many of the world's greatest businesses use it for their profession or company.

Global Popularity



It remains the most popular programming language for developers and back-end developers throughout the world

Versatile Applications



C is used in all hardware and software technologies, making it advantageous for both aspiring and seasoned software specialists

Project Development



As a developer, think outside the box, devise imaginative solutions using available resources, and contribute to the future of software



Thank you for your attention!

